

Sreevedh Jella

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PROFILE

Final-year Computer Science (AI & ML) student with hands-on experience building and testing software systems using Python, SQL, REST APIs, and backend frameworks. Strong foundation in data structures, algorithms, OOP, databases, and system design, with experience developing AI-enabled applications, debugging complex pipelines, and optimizing system performance.

EXPERIENCE

Segritech (Tikkly Agro Solutions Private Limited)

Sept 2025 - Mar 2026

Software Development Intern

- Contributed to an AI-based crop quality inspection system using Python and OpenCV, building defect detection models that processed 5,000+ produce images and achieved 91% grading accuracy, supporting the company's core agri tech product line.
- Built a web-scraping pipeline in Python to collect nationwide temple data across 12,000+ locations for a founder-led side project, structuring the data for a Rapido style booking platform, and prototyped the product UI in Figma.

SKILLS

Languages: Python, SQL, Java

Computer Science: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Software Engineering: REST APIs, FastAPI, Git, Docker, Automated Testing, Debugging

Databases: PostgreSQL, Supabase

AI/ML: Machine Learning, Deep Learning, NLP, RAG, LLMs, LlamaIndex, OpenCV, OCR

EDUCATION

BVRIT, Narsapur

2023 - 2027

B.Tech, Computer Science Engineering (AI & ML)

CGPA: 8.40/10

ACHIEVEMENTS

- LeetCode: 250+ problems solved.
- Sreevedh Jella et al., "Hybrid Motion Synthesis for Speech-to-Sign Translation using Deterministic Linguistic Parsing & GRUs," *I3CTCON 2026*, Mar. 2026. IEEE Xplore. DOI: [10.1109/I3CTCON68242.2026.11507164](https://doi.org/10.1109/I3CTCON68242.2026.11507164)
- Placed 4th out of 240 teams (2nd in domain) at Demux 2.0, a 24 hrs national-level hackathon for MediscanAI.
- Led Promethean'25 for Dark Auction; served as School Sports Captain (25+ medals).

PROJECTS

AI-powered Repository Intelligence Platform | *LlamaIndex, BM25, ChromaDB, ColBERT, FastAPI, Supabase*



- Built CodeBase-Copilot, a full-stack platform that indexes and analyzes engineering repositories, delivering 12 automated workflows for architecture insights, security review, and technical-debt analysis through a hybrid retrieval engine.
- Engineered and profiled a hybrid retrieval pipeline combining dense search, BM25, RRF, and Cross-Encoder reranking, achieving 0.80+ Hit Rate, 0.60+ MRR, and ~0.1s baseline retrieval latency across evaluation queries.
- Built a 238-test automated evaluation suite covering routing, authentication, persistence, API behavior, and retrieval workflows, with evaluation of Hit Rate, MRR, faithfulness, relevancy, and latency.

MediScanAI: A Privacy-First Multimodal AI Health Copilot | *FastAPI, FAISS, BM25, Docker, PostgreSQL, React*



- Built MediScanAI, an end-to-end multimodal application processing text, voice, and medicine images through OCR, speech recognition, hybrid retrieval, and local LLM inference.
- Engineered a 4-stage retrieval pipeline combining FAISS semantic search, BM25 keyword matching, Reciprocal Rank Fusion, and Cross-Encoder reranking, with PostgreSQL-backed persistence for users, analyses, and audit logs.
- Hardened the system with JWT authentication, bcrypt password hashing, input validation, rate limiting, bounded ML concurrency, prompt-injection defenses, Docker deployment, and automated unit/API/database/E2E tests.

FedSegX - Federated Cross Domain Segmentation | *PyTorch, FedProx, PVTv2*



- Built a distributed federated learning system where two clients collaboratively trained a shared model without exchanging raw data, using FedProx aggregation for cross-domain learning.
- Diagnosed catastrophic cross-domain collapse in single-domain baselines, with Dice scores dropping from 0.97 to 0.15 and 0.82 to 0.22, then recovered to 0.92 and 0.80 Dice via FedProx aggregation.
- Designed a dual-head PVTv2-B2 architecture with auxiliary edge supervision and a manifest-tracked data-split protocol, guaranteeing zero data leakage across 50 federated training rounds.